SPPA-MVFit Submission Supplement: Protocol, Sealed Results, and Claim Boundaries

Purpose

This short supplement is the formal companion for the SPPA-MVFit paper. The longer historical technical diary is **not** submitted. This document records the sealed confirmatory protocol path, key hashes, and claim boundaries. Primary scientific tables appear in the main paper.

Primary endpoint (held-out)

- Estimand: clean 64-cubed voxel IoU of `sppa_mvfit` minus `generic_mvfit`.
- Units: 240 source actors (12 family$\times$stratum cells, equal weight).
- Decision: stratified bootstrap 95% CI lower bound $> +0.030$.
- Result: mean $\Delta = 0.190$, CI $[0.181, 0.199]$, **H1 PASS**.
- Strata: CSG-ID 0.209, implicit-OOD 0.172 (both positive).
- Provenance: `synthetic_geometry` only.

Reproduction commands

Run from the git repository root that contains the `paper_semantic_proxy_3d` junction (or paper directory). Do not re-optimize the sealed held-out methods after inspection.

Table 1: Minimal checks for the submitted primary claim.

Command	Purpose	Expected
`python-mpytestpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/tests/test_contract.py`	Shared builder, budget, no cross-imports, split balance.	7 passed
`pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/verify_package.py--development`	Development package integrity.	PASS
`pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/export_paper_tables.py`	Regenerate main-paper tables from sealed `raw_metrics.csv`.	writes `.tex`
`pythonpaper_semantic_proxy_3d/tools/reproduce_sppa_mvfit_paper.py--strict`	Hash/gate checks for the submission snapshot.	0 blockers

Held-out generation/evaluation commands exist in `reproducibility/sppa_mvfit/README.md` and require the protocol PASS record, pretest freeze, and NIST-bound seed manifest. Re-running method optimization for a better score is forbidden by protocol.

Selected sealed hashes

- Pretest freeze SHA-256: `8E2ADBF32F299B24CD2A5AB87C74D142E707696F79D18DF0C3332209C3B46CA3`
- Protocol PASS SHA-256: `2348946BDDB04B8E5CA7D2C845C5F5C45F1AE06F8907E99218ED5E9A379FA74F`
- Sealed predictions binary SHA-256: `F870C57D9CC6FF4868EFB25FD2926FA7D19858EAF8CD0E9781F38990D7D145FD`
- Raw metrics SHA-256: `57A82D234F55013D76BEF2E36CF2B3F7C5617DD4FA6EF811C2A8447A04C0AD63`

Claim boundaries

Supported: family-conditioned occupancy gain under synthetic calibrated silhouettes; shared generator; optional runtime contract evidence.

Not supported: measured flight, real-UAV 3D GT accuracy, operator benefit, universal open-set geometry, or neural image-to-3D SOTA ranking.

Real-image figures in the main paper remain qualitative probes without 3D ground truth and use declared replay assumptions where telemetry is shown.